Nate Annau

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Applied Mathematics Student interested in Computational Science

Education

Technical University of Munich
Master of Science, Mathematics in Science and Engineering
Oct 2026 — August 2028 (expected) Munich, Germany

University of California, Santa Barbara
Bachelor of Science, Mathematics
Bachelor of Science, Computer Science
Sep 2022 — Jun 2026 Santa Barbara, California

- Graduated with Honors (3.88 GPA)
- Notable Coursework: Real, Numerical, and Complex Analysis; Linear Algebra; Data Structures & Algorithms; Machine Learning; Functional Programming; Computer Security
- Performed undergraduate research in the Computational Applied Science Lab, developing a CUDA-based solver for Hamilton-Jacobi Equations
- Completed a Directed Reading and associated poster presentation, covering Coxeter Groups and Buildings

Experience

Research and Development Intern
Sandia National Laboratories
Jun 2026 — Aug 2026 Livermore, California

- Performed classified formal methods work in the `Rocq` language under the direction of a laboratory mathematician
- Took part in the AI agents for Scientific Workflows Challenge at nearby Lawrence Livermore National Labs

Software Engineering Intern
Playlab AI
Jun 2025 — Sep 2025 Remote

- Developed custom benchmarking framework to statistically evaluate 10+ document parsing services across production documents, identifying optimal parsing strategies for RAG
- Expanded parsing and chunking capabilities significantly in the internal API, adding handling for common data formats
- Used heavy typing features in Python, focusing on maintainability and readability, and greatly expanded unit tests

Software Engineering Intern
HiveTop
Jun 2024 — Aug 2024 Berlin, Germany

- Developed key pages, including a graph dashboard for device telemetry, for a frontend platform using `Vue` and `Tailwind`
- Created REST endpoints in the `Rust` backend to facilitate data transmission from embedded devices
- Architected major elements of our CI / CD setup, using Github actions, `ESLint`, and `mypy`

Projects

Ergo

- In collaboration with my friend, developed a `Typst` package for notetaking and assignments for Math, CS, and Physics students
- The package is actively maintained, has 50+ stars on Github, and is being used for a PhD thesis at a French university
- Contains over a dozen custom environments, custom color schemes, and over one hundred macros for common symbols

Cybersecurity Competitions

- Won second place out of 300+ participants in the Geo OSINT challenge at DEF CON, the world's largest hacker conference
- As the second highest scoring individual contributor on my CTF team, placed in the top 6% in the PwnMe CTF Qualifier
- Captained a team of 10 at iCTF 2026, leading the team to a 15th place finish

Courses Graph

- Scraped the UCSB website with `Python`, queried the UCSB API, and synthesized the data as JSON
- Dynamically displayed course data using a constrained optimization library and `d3`, with `Quasar` components
- Presented working prototype in front of audience of peers and industry judges at the 2023 Coders SB Project Showcase, winning third place out of ~35 projects
- At SBHacks 2025, extended this general concept to a visualization of the dependency graph of Proof Wiki

Open Source Contributions

- Contributor to `weo-reader`, a Python client to read world economic data from the IMF as a `pandas` dataframe
- Authored just under one thousand Wikipedia contributions, and wrote several tools to help automate data collection
- Made corrections and wrote a chapter for a popular online summary textbook of early undergraduate math courses

Languages and Tools

Python (numpy, pandas, Flask)
Git
Linux (Ubuntu, Arch, CentOS, Nix)
C++ and CUDA
TypeScript (Vue.js, Svelte)
Neovim
Rust
Interactive Theorem Provers (Rocq, Lean)